A Hybrid Decomposition based approach for Quantifying green house gas emissions

Pallavi Gajavalli[1], Muskan Pathan[1], Poojitha Makkapati[1], Prakriti Sudha[1] and Naba Krushna Sabat[1*]

[1*]Department of Electronics and Communication Engineering, Vignan's Foundation for Science, Technology and Research, Vadlamudi, Guntur, 522213, Andhra Pradesh, India.

*Corresponding author(s). E-mail(s): nks_ece@vignan.ac.in;
Contributing authors: gajavalli2004@gmail.com; muskan0211@gmail.com; mpoojitha985@gmail.com; sudaprakruthi@gmail.com;

Abstract

The emission of greenhouse gases (GHG) has contributed greatly to climate change and environmental degradation and therefore proper forecasting is a key aspect of sustainable development and policy planning. Nonlinearity, non-stationarity, and external effects however make real-world emission data very complex, which restricts the usefulness of conventional prediction models. In order to overcome these difficulties, this article suggests a hybrid decomposition-based methodology, which combines Variational Mode Decomposition (VMD), Autoregressive Integrated Moving Average (ARIMA), and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models. This article breaks down the original emission time-series into intrinsic mode functions, and allows simplified modeling of complex patterns. ARIMA fits the linear trends and the time-dependent characteristics whereas the volatility and the uncertainty of the data is captured using GARCH. The proposed model is experimented with reference to the actual data sets of the regions like India and Japan which are the real life scenario of emissions. The model performance is evaluated using standard evaluation measures like Mean Absolute error (MAE), Mean Square error(MSE), Root Mean Square error (RMSE) and R^2 score. The results point to the fact that the proposed hybrid model ARIMA-VMD-GARCH is more effective and appropriate when compared to the conventional models.

Keywords: Greenhouse Gas Emissions, Time Series Forecasting, ARIMA, VMD, GARCH, Hybrid Model, Climate Change, Environmental Monitoring

1 Introduction

Greenhouse gas (GHG) emissions play a significant role in global climate change and have a major impact on environmental sustainability and human livelihoods [9], [15]. Accurate monitoring and forecasting of these emissions are essential for effective climate policy formulation, environmental management, and sustainable development planning [16],[1]. However, GHG emission trends are highly complex and influenced by multiple factors such as industrial activities, energy consumption, and seasonal variations, making accurate prediction a challenging task [29].

Traditional statistical models such as Autoregressive Integrated Moving Average (ARIMA) and Seasonal ARIMA (SARIMA) are widely used for time series forecasting [8],[3],[27]. These models are effective in capturing linear trends and seasonality; however, real-world emission data is often nonlinear, non-stationary, and volatile. As a result, these models may fail to accurately represent complex variations in the data[23],[26]. This limitation can lead to reduced forecasting accuracy, thereby affecting reliable decision-making in climate mitigation and environmental planning. To address these challenges,researchers have explored advanced approaches such as decomposition-based and hybrid models [10, 11, 18, 24]. VMD has gained attention due to its ability to decompose complex time series data into simpler Intrinsic Mode Functions (IMFs), thereby reducing data complexity and improving modelling efficiency [5, 13, 14, 25] Similarly, volatility model such as GARCH is used to capture time-varying variance and uncertainty in the data [2, 6, 12, 28]. However, using these models individually is often insufficient to capture all the characteristics of emission data.

Effective forecasting of GHG emissions requires modelling both trend and variability in the data. Emission patterns vary significantly across regions such as India and Japan, indicating the need for flexible and adaptive forecasting models [31]. These variations can significantly influence environmental policies and sustainability goals, highlighting the importance of accurate prediction systems.

With increasing climate uncertainty and the growing need for reliable environmental analysis, there is a strong demand for intelligent and integrated forecasting frameworks [15]. Therefore, this study proposes a hybrid decomposition-based model that combines VMD, ARIMA, and GARCH to improve prediction accuracy and robustness. The proposed framework provides a reliable and scalable solution for GHG emission forecasting and supports evidence-based decision-making in climate policy and sustainable development.

2 Contributions

The following are the major contributions of this article:

- A novel hybrid forecasting framework based on VMD–ARIMA–GARCH is proposed for accurate greenhouse gas (GHG) emission prediction by effectively handling the nonlinear and non-stationary characteristics of emission data.
- Experimental results demonstrate that the proposed VMD–ARIMA–GARCH model outperforms conventional forecasting models, including ARIMA, SARIMA, ARIMAX, SARIMAX, VMD–ARIMA, and VMD–SARIMA, using standard statistical evaluation metrics.
- The integration of Variational Mode Decomposition (VMD) improves signal decomposition and feature extraction by separating complex GHG emission signals into Intrinsic Mode Functions (IMFs), thereby enhancing prediction accuracy.
- The effectiveness and robustness of the proposed approach are validated using greenhouse gas datasets from India and Japan, demonstrating its applicability for multi-region environmental forecasting.
- The proposed forecasting framework can support environmental monitoring and policy planning by providing reliable long-term GHG emission predictions for sustainable development initiatives.

3 Organization

The remainder of this article is organized as follows: Section 4 discusses the related literature on greenhouse gas emission forecasting. Section 5 explains the problem formulation and objectives of the study. Section 6 describes the proposed methodology. Section 7 presents the experimental setup, evaluation metrics, performance metrics, model training, comparative analysis, result and discussion through subsections. Finally, Section 8 concludes the article, followed by Author Contributions and Availability of Datasets sections.

4 Related work

The study of greenhouse gas (GHG) emission forecasting has gained significant attention in recent years due to increasing concerns about climate change and the need for sustainable environmental management [15, 16]. Accurate prediction of emission trends is essential for policy formulation and achieving global climate targets. However, GHG emission data is influenced by multiple environmental and socio-economic factors, making it highly complex and challenging to model [12].

4.1 Traditional Statistical Models for GHG Forecasting

Traditional statistical approaches such as ARIMA and SARIMA are widely used due to their simplicity and effectiveness in capturing linear trends and seasonality [8]. However, these models face several limitations when applied to real-world emission data such as

- Inability to capture nonlinearity [3, 23]
- Sensitivity to non-stationarity
- Limited capability in modelling complex patterns

These limitations restrict the effectiveness of standalone statistical methods for accurate emission forecasting.

4.2 Decomposition-Based Forecasting Approaches

To address the complexity of time series data, decomposition-based methods such as Variational Mode Decomposition (VMD) have been introduced. VMD decomposes a complex time series into multiple Intrinsic Mode Functions (IMFs), each representing a simpler frequency component [5]. This decomposition improves modelling efficiency by reducing noise and non-stationarity.

Studies such as Lahmiri [14] have shown that combining VMD with forecasting models like ARIMA enhances prediction accuracy by simplifying the underlying data structure. However, decomposition-based methods also have certain limitations:

- **Parameter sensitivity:** The performance of VMD depends on the selection of decomposition parameters.
- **Lack of standalone predictive capability:** VMD only decomposes data and must be integrated with forecasting models.
- **Computational overhead:** The decomposition process increases computational complexity, especially for large datasets.

4.3 Volatility Modelling in Emission Data

GHG emission data often exhibits time-varying variance and irregular fluctuations. Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models are widely used to capture such volatility patterns [2, 6, 12, 28]. These models are effective in modelling uncertainty and variance clustering in time series data.

However, GARCH models also have inherent limitations:

- **Focus on variance rather than mean behaviour:** They do not model the overall trend of the data.
- **Dependence on residuals:** GARCH requires residuals from another model, making it dependent on prior modelling steps.
- **Limited standalone forecasting capability:** It cannot provide complete predictions independently.

4.4 Hybrid Models and Research Gap

To overcome the limitations of individual models, recent research has focused on hybrid approaches[25] that integrate decomposition, statistical modelling, and volatility analysis [15, 28]. For instance, Li et al [14] demonstrated that a VMD–ARIMA model improves forecasting accuracy by handling non-stationarity, while Zhang et al. [30] incorporated GARCH to capture volatility in time series data. These studies highlight the effectiveness of combining multiple techniques to enhance prediction performance.

Despite these advancements, several limitations remain:

- **Lack of fully integrated frameworks:** Many models do not simultaneously capture trend, nonlinearity, and volatility within a single system [27].
- **Inadequate handling of complex data characteristics:** Existing approaches struggle to model nonlinear, non-stationary, and volatile behaviours together[7].
- **Limited adaptability across regions:** Models often fail to generalize across datasets from different geographical regions such as India and Japan.

4.5 Need for the Proposed Hybrid Model

From the above analysis, it is evident that existing forecasting methods are either insufficient in capturing the complex characteristics of GHG emission data or lack integration of multiple modelling techniques. To address these limitations, this study proposes a hybrid framework that:

- Decomposes complex time series into IMFs using VMD,
- Models linear temporal dependencies using ARIMA on each IMF,
- Captures volatility and uncertainty using GARCH on residuals,
- Improves prediction accuracy across diverse regional datasets.

Recent studies [12, 24] further emphasize the need for such integrated hybrid approaches. The proposed framework bridges the gap between traditional statistical models and advanced hybrid techniques, providing a robust and scalable solution for GHG emission forecasting.

5 Problem formulation

Let $X = \{x_1, x_2, \ldots, x_n\}$ denote a univariate time series of greenhouse gas (GHG) emissions, where x_t represents the observed emission value at time step t. The objective is to forecast future values $\hat{x}_{t+h}$ using historical observations. GHG emission data exhibits complex characteristics such as nonlinearity, non-stationarity, and time-varying volatility, which limit the effectiveness of conventional forecasting models and reduce prediction accuracy. Let $F(\cdot;\theta)$ denote a forecasting model with parameters θ, where the prediction at time t is given by $\hat{x}_t = F(x_1, x_2, \ldots, x_{t-1})$. The model parameters are estimated by minimizing the empirical loss defined as

$$\min_{\theta} \frac{1}{n} \sum_{t=1}^{n} L(\hat{x}_t, x_t) \tag{1}$$

where x_t is the actual observed value, $\hat{x}_t$ is the predicted value, $L(\cdot)$ is a suitable loss function (such as Mean Squared Error or Mean Absolute Error), and n is the total number of observations. The goal is to develop a robust forecasting framework capable of effectively capturing the complex temporal dynamics of GHG emission data to improve prediction accuracy.

6 Proposed approach

A hybrid decomposition model, termed the ARIMA-VMD-GARCH model, is proposed for the prediction of greenhouse gas (GHG) emissions using historical time series data, with the procedural steps outlined below.

6.1 Data acquisition and study regions

Two geographical regions— India and Japan—are selected as the study areas in this research. The annual greenhouse gas emission data (including land use) for India and Japan were obtained from an open-source climate data repository [20], covering the period from 1850 to 2024 with 175 samples. All datasets contain the scaled total annual GHG emissions, normalized to enable cross-regional comparison. These long-term historical records provide a rich and diverse basis for prediction model development and validation.

6.2 Data Preprocessing

The greenhouse gas (GHG) emission data collected for countries such as India and Japan is first prepared to make it suitable for model development. Only the required features, namely the year and corresponding emission values, are selected, and the data is arranged in chronological order to preserve its sequence.

In the initial stage, any incomplete or inconsistent records are removed to avoid errors during analysis. The cleaned dataset is then divided into three parts: 70% for training, 20% for validation, and 10% for testing. This split allows the model to be trained effectively while also being evaluated on unseen data.

To further improve model performance, Min-Max scaling is applied to the data. This transformation brings all values into a similar range, helping the model learn patterns more efficiently without being affected by large variations in magnitude. The scaling parameters are learned from the training data and then applied consistently to the validation and test datasets.

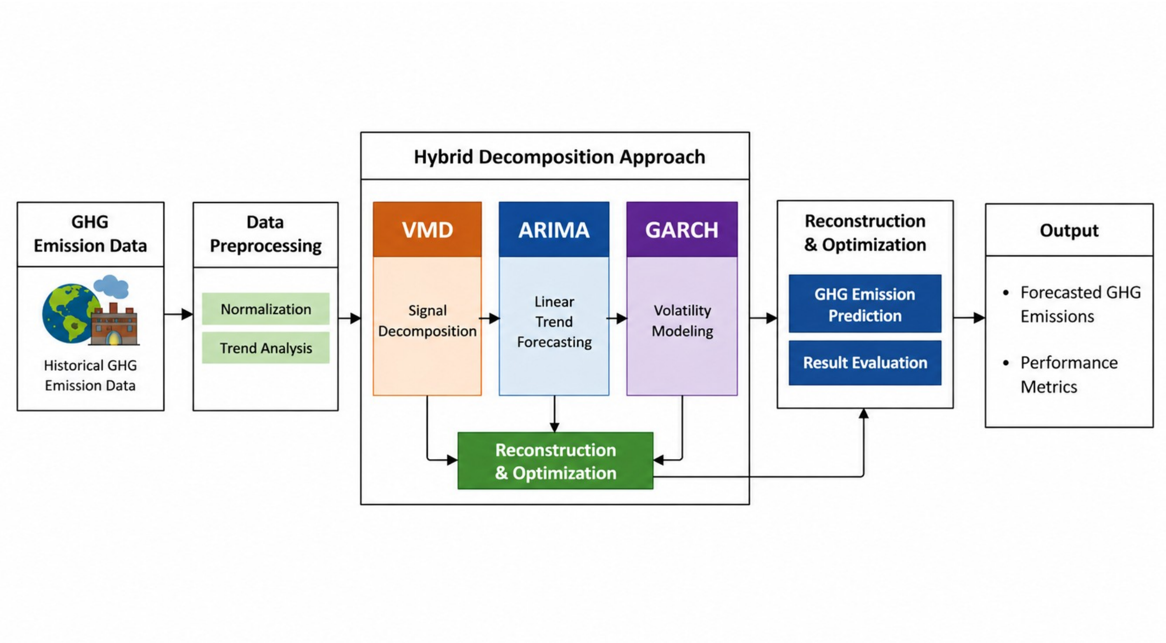

Fig. 1: Block schematic of the hybrid decomposition model for predicting greenhouse gas emissions.

After scaling, the data is processed using Variational Mode Decomposition (VMD), which separates the original emission signal into multiple simpler components. Each component represents different underlying patterns in the data, making it easier for the model to capture important features.

Overall, these preprocessing steps convert the raw data into a structured and reliable form, which enhances the performance of the proposed model for GHG emission prediction.

6.3 Variational Mode Decomposition (VMD)

The internal working of the VMD module is illustrated in Fig. 2. The raw GHG emission signal is initially pre-processed through normalisation and mirror-padding to reduce boundary distortions at the edges of the time series. The signal is then transformed into the frequency domain to capture its spectral characteristics.

Variational Mode Decomposition aims to decompose the input signal $x(t)$ into a set of K band-limited intrinsic mode functions (IMFs), such that:

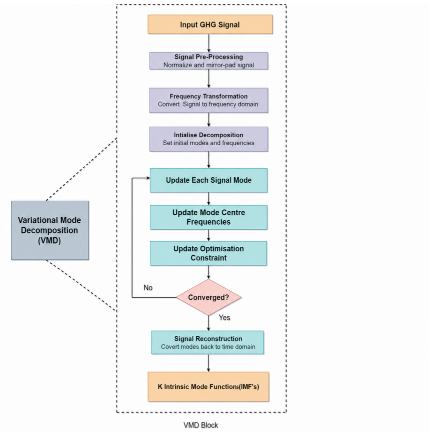

Fig. 2: Block diagram of Variational Mode Decomposition (VMD).

$$x(t) = \sum_{k=1}^{K} u_k(t) \tag{2}$$

where $u_k(t)$ denotes the k^{th} mode.

The decomposition is achieved by solving a constrained variational problem, where the bandwidth of each mode is minimised subject to the constraint that all modes sum to reconstruct the original signal.

An iterative optimisation process based on the Alternating Direction Method of Multipliers (ADMM) is employed to solve this problem. In each iteration, each signal mode is updated to best represent its assigned frequency band, the centre frequency of every mode is adjusted toward the dominant energy concentration of that mode, and the optimisation constraint is tightened. These steps are repeated until the relative change between successive iterations falls below a predefined convergence threshold [19].

Finally, the decomposed modes are reconstructed back into the time domain, resulting in K Intrinsic Mode Functions (IMFs). These IMFs represent the original signal at different frequency scales and serve as simplified, individually modelable components for subsequent forecasting.

6.4 ARIMA Modelling

The ARIMA model, illustrated in Fig. 3, is applied independently to each IMF $u_k(t)$ obtained from the VMD module. The process begins by defining a candidate set of model orders (p, d, q), where each parameter is varied within a predefined range.

For each IMF component $u_k(t)$, an ARIMA model is fitted to capture its linear temporal dynamics, which can be expressed as: [21]

$$\phi_p(B)(1-B)^d u_k(t) = \theta_q(B)\epsilon_t \tag{3}$$

where B is the backshift operator and ϵ_t represents the residual error.

A grid search strategy combined with walk-forward validation is employed to identify the optimal model parameters. Specifically, the model is trained iteratively on an expanding window of historical data, generating one-step-ahead forecasts at each step. The prediction error is computed for each candidate configuration, and the model with the lowest Root Mean Square Error (RMSE) is selected as the best-performing set of orders (p^*, d^*, q^*) [4].

Using these optimal orders, final forecasts are generated for both the validation and test periods. The predictions obtained from all K IMF components are then aggregated by summation to reconstruct the overall GHG emission forecast. The residuals from this stage defined as the difference between actual and predicted values are subsequently forwarded to the GARCH model for volatility modelling.

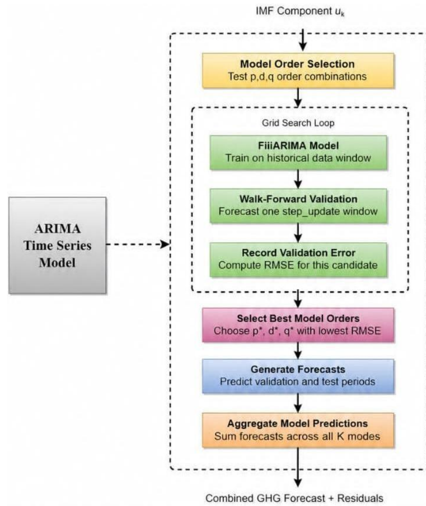

Fig. 3: Block diagram of ARIMA modelling with walk-forward grid search.

6.5 GARCH Volatility Modelling

The GARCH stage, illustrated in Fig. 4, is employed to model the volatility and uncertainty present in the residuals obtained from the ARIMA predictions. The residuals, defined as the difference between the observed and predicted values on the validation set, are first extracted and analysed [17].

A GARCH$(1, 1)$ model is fitted to the residual series to capture the time-varying variance, which can be expressed as:

$$\sigma_t^2 = \alpha_0 + \alpha_1\epsilon_{t-1}^2 + \beta_1\sigma_{t-1}^2 \tag{4}$$

where σ_t^2 represents the conditional variance and ϵ_t denotes the residual error at time t.

A stability condition is verified to ensure that:

$$\alpha_1 + \beta_1 < 1 \tag{5}$$

which guarantees stationarity of the variance process. The fitted GARCH model is then used to estimate and forecast the conditional volatility over the test period.

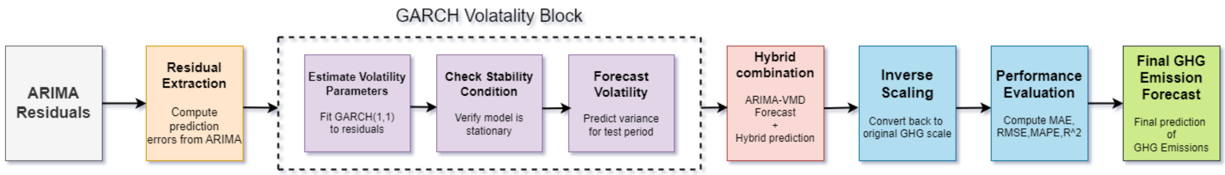

Fig. 4: Architecture of GARCH volatility modelling and hybrid combination.

The ARIMA–VMD framework provides the mean forecast, while the GARCH model characterises the associated uncertainty in the predictions. These two components are combined at the addition node, so that the final output jointly captures both the predicted trend and the predicted volatility.

6.6 Performance Metrics

In this study, the performance of the proposed GHG emission prediction model is evaluated using mathematically defined error metrics, commonly cited as standard statistical measures for forecasting accuracy assessment, which quantify the forecasting error and prediction accuracy between the actual and predicted emission values. These are the mathematically defined error metrics used for forecasting performance evaluation [22]:

$$MAE = \frac{1}{M}\sum_{i=1}^{M}|\varepsilon_i| \tag{6}$$

$$MSE = \frac{1}{M}\sum_{i=1}^{M}(\varepsilon_i)^2 \tag{7}$$

$$RMSE = \sqrt{\frac{1}{M}\sum_{i=1}^{M}(\varepsilon_i)^2} \tag{8}$$

$$R^2 = 1 - \frac{\sum(A_i - \hat{P}_i)^2}{\sum(A_i - \bar{A})^2} \tag{9}$$

Here, ε_i represents the difference between the actual emission value A_i and the predicted value $\hat{P}_i$, i.e., $\varepsilon_i = A_i - \hat{P}_i$. The term M denotes the total number of observations, and $\bar{A}$ represents the mean of the actual emission values.

7 Experimental Analysis

The performance of the proposed model is analyzed through a series of experiments, which are explained in detail in the following sections.

7.1 Experimental Setup

The configuration used to carry out the experiments for the proposed model is summarized in Table 1.

The proposed hybrid approach combining ARIMA, VMD and GARCH is implemented using the setup. The datasets for India and Japan are preprocessed and divided into training, validation, and testing subsets. VMD is used to decompose the emission data into multiple components, after which ARIMA is applied to model each component individually. GARCH is then used to handle variations in the residuals. The final performance of the model is evaluated using standard error metrics.

7.2 Model Training

The proposed hybrid model for greenhouse gas (GHG) emission prediction is based on the integration of VMD, ARIMA, and GARCH techniques. Initially, the normalized emission data is decomposed into multiple components using VMD with $K = 3$, which helps in breaking down the complex signal into simpler modes. Each of these components is then modeled individually using the ARIMA approach.

For ARIMA modeling, a grid search method is used to determine the optimal parameters (p, d, q) for each mode. A walk-forward validation strategy is applied during training, where the model is updated step-by-step using new observations. This approach helps in simulating real-time prediction conditions. If the ARIMA model fails to converge, the last observed value is used as a fallback prediction.

After generating predictions from all modes, the outputs are combined to obtain the final prediction. To further improve accuracy, residual errors from the validation phase are analyzed, and a GARCH-based adjustment is applied to capture remaining variations.

Table 1: Experimental Setup

Hardware Components	Specification	Software Components	Specification
CPU	Intel Core i5/i7 (64-bit)	Operating System	Windows 10 / Linux
RAM	8 GB or higher	Programming Language	Python 3.9
Storage	512 GB / 1 TB	Libraries Used	NumPy, Pandas, Matplotlib, Scikit-learn, Statsmodels
		Modeling Techniques	ARIMA, VMD, GARCH
		Evaluation Metrics	MAE, RMSE, R^2, MAPE

Algorithm 1 Hybrid VMD–ARIMA–GARCH for GHG Emission Forecasting

Require: Time series data $X = \{x_1, x_2, \dots, x_n\}$, Ground truth labels Y
Ensure: Predicted GHG emissions $\hat{X}(t)$
  Data Pre-processing
  Load GHG emission dataset and sort chronologically
  Split dataset into training, validation, and test sets (70:20:10)
  Apply Min–Max normalization using training data:
  $X_{\text{scaled}} = \text{MinMaxScaler}(X_{\text{train}})$
  Variational Mode Decomposition (VMD)
  Decompose signal into K IMFs:
  $\{u_1(t), \dots, u_K(t)\} = \text{VMD}(X_{\text{scaled}})$
  ARIMA Modelling
  Define $(p, d, q) \in \{0, 1\}$
  for each IMF $u_k(t)$ **do**
    Train ARIMA models
    Perform walk-forward validation
    Compute RMSE
    Select best (p^*, d^*, q^*)
    $\hat{u}_k(t) = \text{ARIMA\_Forecast}(u_k(t), p^*, d^*, q^*)$
  end for
  $\hat{X}_{\text{ARIMA}}(t) = \sum_{k=1}^{K} \hat{u}_k(t)$
  $\epsilon_t = Y(t) - \hat{X}_{\text{ARIMA}}(t)$
  GARCH Modelling
  $\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$
  Hybrid Forecast
  $\hat{X}(t) = \hat{X}_{\text{ARIMA}}(t) + \sigma_t$
  Post-processing
  $\hat{X}_{\text{final}} = \text{InverseScaler}(\hat{X}(t))$
  Evaluate using MAE, RMSE, R^2
  return $\hat{X}_{\text{final}}$

The dataset is divided into training, validation, and testing sets in the ratio of 70%:20%:10%. The performance of the model is evaluated using metrics such as MAE, MSE, RMSE and R^2 Score.

The results show that the proposed model produces stable and accurate predictions for GHG emission data.

7.3 Comparative Analysis

The comparative performance analysis of different forecasting models for the India and Japan datasets is presented in Tables ?? and ??. The proposed hybrid model was compared with several conventional and intermediate models, including ARIMA, SARIMA, ARIMAX, SARIMAX, ARIMA–VMD, SARIMA–VMD, and SARIMA–VMD–GARCH. The evaluation was performed using standard statistical metrics such as MAE, MSE, RMSE and R^2 score.

For the India dataset, traditional statistical models such as ARIMA, SARIMA, ARIMAX, and SARIMAX produced satisfactory forecasting results with moderate error values and R^2 scores above 0.89. The incorporation of VMD improved the ability of the models to capture hidden patterns in the emission data, leading to better forecasting performance. Further enhancement was observed after integrating the GARCH component, which effectively handled volatility and uncertainty in the residuals. Among all the compared models, the proposed hybrid model achieved the best overall performance with the lowest MAE Value of 0.0176, MSE value of 0.0005, RMSE of 0.0215, and the highest R^2 score of 0.9679, demonstrating its effectiveness in modeling both trend and variability in GHG emissions.

Similarly, for the Japan dataset, the baseline forecasting models achieved comparatively good results. Although the inclusion of VMD improved prediction capability to some extent, the addition of GARCH significantly enhanced the forecasting accuracy by capturing fluctuations present in the time-series data. The proposed hybrid model outperformed all other models by achieving the lowest MAE value of 0.0122, MSE value of 0.0003 and RMSE value of 0.0147 along with the highest R^2

score of 0.9660. These results confirm the capability of the proposed hybrid model to effectively forecast complex and volatile emission patterns.

Overall, the experimental analysis demonstrates that hybrid approaches provide superior forecasting performance compared to standalone statistical models. The integration of VMD simplifies complex emission signals, while GARCH effectively captures residual volatility, resulting in improved prediction accuracy and robustness. Therefore, the proposed hybrid model can be considered a reliable and efficient framework for greenhouse gas emission forecasting.

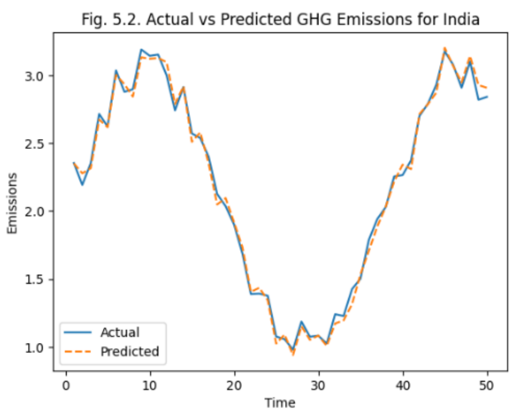

Fig. 5: Actual vs Predicted GHG Emissions for India

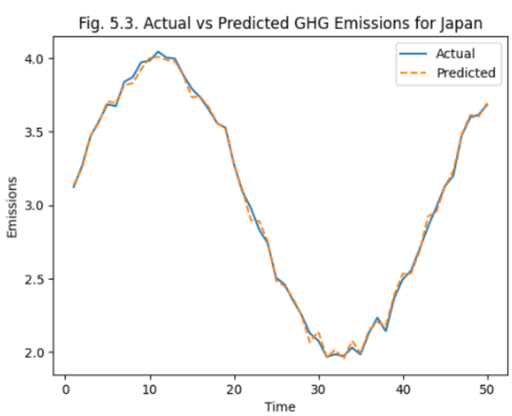

Fig. 6: Actual vs Predicted GHG Emissions for Japan

8 Conclusion

This study presented a hybrid decomposition-based framework for forecasting greenhouse gas (GHG) emissions by integrating VMD, ARIMA, and GARCH models. The primary objective was to address the challenges associated with non-linearity, non-stationarity, and volatility in real-world emission data, which often limit the performance of traditional forecasting approaches.

The experimental results demonstrate that the proposed hybrid model significantly improves prediction accuracy by effectively capturing both the underlying trends and the variability present in the data. VMD plays a crucial role in decomposing complex time-series signals into simpler components, thereby enabling more precise modeling. ARIMA effectively captures temporal dependencies, while the GARCH component enhances the model by accounting for time-varying volatility and uncertainty.

Across two datasets, India, and Japan—the proposed model consistently outperformed traditional and partially hybrid models by achieving lower error metrics and higher R^2 scores. In the dataset, where the data is highly irregular and volatile, the model demonstrated stable and reliable performance. In contrast, for the India and Japan datasets, which exhibit more structured patterns, the model achieved very high predictive accuracy, confirming its adaptability across different data characteristics.

Overall, the findings clearly indicate that hybrid approaches are more effective than standalone models for complex time-series forecasting tasks. The integration of decomposition and volatility modeling enables a more comprehensive understanding of both the structural and stochastic components of the data. Among all evaluated models, ARIMA-VMD-GARCH emerged as the most robust and reliable approach for GHG emission prediction.

In conclusion, the proposed framework provides a scalable and efficient solution for accurate GHG emission forecasting. It has strong potential to support data-driven decision-making in climate policy, environmental management, and sustainable development. Furthermore, this approach can be extended to other domains involving complex and non-stationary time-series data, making

Table 2: Comparative Performance Analysis of Different Models for India Dataset

MODELS	MAE	MSE	RMSE	R^2 Score
ARIMA	0.0318	0.0015	0.0390	0.8953
SARIMA	0.0275	0.0015	0.0386	0.8972
ARIMAX	0.0287	0.0014	0.0368	0.9065
SARIMAX	0.0273	0.0012	0.0353	0.9012
ARIMA - VMD	0.1540	0.0310	0.1761	0.9243
SARIMA - VMD	0.1151	0.0220	0.1483	0.9463
SARIMA - VMD - GARCH	0.0159	0.0006	0.0251	0.9563
ARIMA - VMD - GARCH	**0.0176**	**0.0005**	**0.0215**	**0.9679**

Table 3: Comparative Performance Analysis of Different Models for Japan Dataset

MODELS	MAE	MSE	RMSE	R^2 Score
ARIMA	0.0225	0.0010	0.0316	0.9124
SARIMA	0.0208	0.0009	0.0300	0.9187
ARIMAX	0.0212	0.0009	0.0298	0.9203
SARIMAX	0.0205	0.0008	0.0289	0.9250
ARIMA - VMD	0.0450	0.0045	0.0671	0.9352
SARIMA - VMD	0.0387	0.0032	0.0566	0.9481
SARIMA - VMD - GARCH	0.0135	0.0004	0.0200	0.9587
ARIMA - VMD - GARCH	**0.0122**	**0.0003**	**0.0147**	**0.9660**

it a valuable contribution to environmental data analytics and forecasting research

Author Contributions

Pallavi Gajavalli contributed to data collection and preparation of the manuscript. Prakriti Sudha and Muskan Pathan were responsible for data preprocessing, model development, and experimental analysis. Poojitha Makkapati contributed to validation, performance evaluation, and visualization of results. Dr. Naba Krushna Sabat supervised the research work, provided conceptual and technical guidance, and performed final review and editing of the manuscript.

Availability of datasets

The datasets used in this study are publicly available on GitHub at: https://github.com/gpallavi-68/GHG-Emissions-Datasets_India-Japan.

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